Disaster Recovery (DR) Solutions Design and Review: Workshop Activity – Seminar Discussion

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Introduction

Cloud computing has transformed organisational information technology by providing scalable, flexible and cost-effective computing resources. However, increasing reliance on cloud service providers introduces challenges relating to vendor lock-in, cybersecurity, data protection and disaster recovery (DR). These challenges can affect an organisation's operational resilience, business continuity and ability to respond effectively to cyber incidents.

Drawing on the work of Kumar (2024) and Corbari et al. (2024), this discussion examines the main vendor lock-in challenges, their mitigation strategies and the security concerns associated with modern cloud environments. It also considers how mission thread analysis can support the design of resilient cloud-based systems.

1. Main Vendor Lock-In Issues and Mitigation Strategies

Vendor lock-in occurs when an organisation becomes excessively dependent on a particular cloud service provider, making it technically difficult, financially expensive or operationally disruptive to migrate to another provider. This dependency can restrict organisational flexibility and increase long-term operational risks (Kumar, 2024).

The main issues include:

1.1 Data portability and interoperability

Cloud providers frequently use proprietary data formats, application programming interfaces (APIs) and storage technologies. These differences can make transferring applications and datasets between platforms difficult. Migration may require significant

system modifications, potentially resulting in operational disruption or data integrity problems.

Mitigation: Organisations should prioritise open standards, interoperable APIs and widely supported data formats. Containerisation technologies such as Docker and orchestration platforms such as Kubernetes can improve application portability, although they do not eliminate dependencies on provider-specific infrastructure (NIST, 2012).

1.2 Financial dependency and migration costs

Vendor lock-in may expose organisations to increased subscription charges, data transfer fees and unexpected pricing changes. Migrating large datasets can also generate substantial expenditure and require specialised technical expertise.

Mitigation: Organisations should undertake total cost of ownership assessments, negotiate transparent service-level agreements (SLAs) and establish contractual exit arrangements. Migration costs and alternative providers should be evaluated before committing to long-term cloud services.

1.3 Proprietary technologies and technical dependencies

Cloud-native applications developed using provider-specific databases, serverless functions and management tools may be difficult to operate on competing platforms. Such dependencies can reduce flexibility and complicate disaster recovery arrangements.

Mitigation: Organisations should adopt modular architectures, infrastructure as code (IaC) and technology abstraction where appropriate. A carefully designed multi-cloud or hybrid-cloud strategy can reduce dependence on a single provider, although this introduces additional management and security complexity.

1.4 Disaster recovery and service availability

Reliance on a single cloud provider may create concentrated operational risk, particularly where critical systems depend on the same infrastructure, geographical region or identity management services. A major outage could interrupt essential business activities.

Mitigation: Organisations should implement geographically separated backups, cross-region replication and, where justified, independent recovery environments. Disaster recovery plans should specify recovery time objectives (RTOs) and recovery point objectives (RPOs), supported by regular restoration and failover testing.

Corbari et al. (2024) provide a complementary perspective through mission thread analysis. Their framework maps essential mission activities to supporting systems, networks and information flows. Applied to cloud disaster recovery, this approach can help organisations identify critical dependencies, prioritise recovery activities and understand how the failure of one service may affect wider operational objectives.

2. Security Concerns in Modern Cloud Environments and Their Mitigation

Modern cloud environments introduce security challenges because organisational data, applications and infrastructure are distributed across interconnected systems, often managed by different service providers.

2.1 Data breaches and unauthorised access

Cloud environments store significant volumes of sensitive information, making them attractive targets for cybercriminals. Weak authentication, excessive user privileges and insecure storage configurations can expose confidential information.

Mitigation: Organisations should implement multi-factor authentication (MFA), role-based access control (RBAC), least-privilege access and encryption of data both at

rest and in transit. Regular security assessments and continuous monitoring can further reduce exposure (Grance and Jansen, 2011).

2.2 Misconfiguration and shared responsibility

Cloud security responsibilities are shared between service providers and customers. Misunderstanding these responsibilities may result in unpatched systems, publicly accessible databases or inadequate security monitoring.

Mitigation: Organisations should establish clear responsibility matrices, conduct regular configuration audits and automate compliance assessments. The National Cyber Security Centre (NCSC, n.d.) emphasises the importance of understanding security responsibilities across different cloud service models.

Cloud infrastructure is vulnerable to phishing, credential theft, ransomware, distributed denial-of-service (DDoS) attacks and exploitation of software vulnerabilities. Successful attacks may compromise essential applications and interrupt organisational operations.

Mitigation: Organisations should implement a zero-trust security approach, endpoint detection and response (EDR), security information and event management (SIEM), vulnerability management and incident response procedures. Immutable backups and regularly tested recovery arrangements can minimise the operational consequences of ransomware.

2.4 Limited visibility and complex system dependencies

Corbari et al. (2024) identify communication gaps between cybersecurity specialists and operational personnel as a significant obstacle to effective cyber defence. When critical systems and their dependencies are poorly understood, organisations may

allocate security resources inefficiently or overlook vulnerabilities affecting essential operations.

Mitigation: Mission Thread Analysis (MTA) can establish a shared understanding of operational requirements and supporting technical infrastructure. Mapping Mission Process Threads (MPTs) and Mission Engineering Threads (MEngTs) helps identify critical assets, communication pathways and system dependencies. This supports risk-based security prioritisation and more effective disaster recovery planning.

Storing and processing information across different geographical locations can create regulatory and privacy challenges. Organisations operating in the United Kingdom must consider applicable data protection requirements, including the UK General Data Protection Regulation (UK GDPR).

Mitigation: Organisations should establish data governance policies, assess international data transfers, maintain appropriate audit trails and select cloud providers capable of meeting relevant regulatory obligations. Data classification and periodic compliance reviews should form part of the cloud security strategy.

3. Critical Reflection

In my view, vendor lock-in and cloud security should be considered together when designing disaster recovery solutions. Although multi-cloud strategies may reduce dependence on individual providers, they can also introduce additional security vulnerabilities through inconsistent configurations, complex identity management and fragmented monitoring.

Therefore, adopting multiple cloud providers does not automatically improve resilience. Organisations must evaluate whether the benefits of portability and redundancy outweigh the additional operational complexity.

Corbari et al. (2024) offer a particularly valuable approach because Mission Thread Analysis connects technical cybersecurity controls with essential organisational activities. For example, an organisation may successfully restore its cloud database following an outage but remain unable to process customer transactions because its authentication service or network connectivity has not recovered.

By identifying these dependencies before an incident, organisations can establish realistic recovery priorities and ensure that disaster recovery strategies support business continuity rather than simply restoring individual technical components.

Conclusion

Vendor lock-in creates significant challenges relating to data portability, proprietary technologies, financial dependency and disaster recovery flexibility. These risks can be mitigated through open standards, modular architectures, appropriate contractual arrangements and carefully evaluated hybrid-cloud or multi-cloud strategies.

Modern cloud security concerns, including data breaches, misconfiguration, cyberattacks and limited operational visibility, require a combination of technical controls, effective governance and continuous risk assessment.

Ultimately, integrating the vendor lock-in considerations discussed by Kumar (2024) with the mission-focused approach of Corbari et al. (2024) can help organisations develop secure, adaptable and resilient cloud-based disaster recovery solutions.

References

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